TITLE PAGE

- **Problem Statement ID – Question 4A**
- **Problem Statement Title – AI-Powered Behavioral Anomaly Detection for Cybersecurity**
- **Theme – Cybersecurity**
- **PS Category – Software**
- **Student Name – Ravi Kishan**

SentinelLens — Explainable Behavioral Anomaly Detection

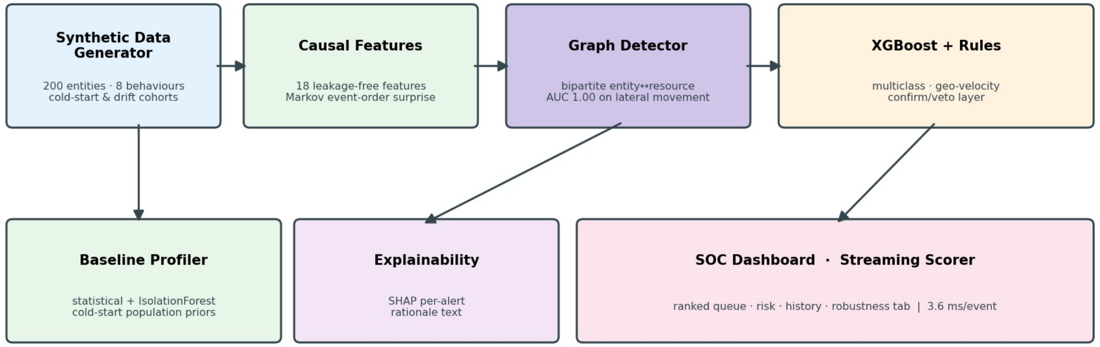
Proposed Solution (Describe your Idea/Solution/Prototype)

- Learns per-entity “normal” behavior (users, service accounts, edge devices) from access logs — then flags deviations in near real-time
- Classifies the attack type — brute force, impossible travel, credential stuffing, lateral movement, device spoofing, low-and-slow exfiltration — not just “anomalous”
- Every alert ships with a SHAP-based plain-English rationale (e.g. “16 accounts from one IP + failed auths”) — built for SOC analysts, not black-box scores
- Addresses signature-based security's blind spot: novel and slow attacks that match no known malware hash or fixed rule
- **Uniqueness: 0% false positives on brand-new entities AND on legitimate behaviour change — the two failures that make real SOC tools unusable**

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware)
- **Python · pandas/NumPy · faker · XGBoost · SHAP · Streamlit · Plotly**
- Generator matches the spec schema and all 8 behaviour patterns — 200 entities, 74K sessions, OS+MAC+protocol fingerprints, no giveaway values
- Sequence-aware detection via GRAPH model (spec option 3): bipartite entity↔resource graph, peer affinity + 2-hop reachability — AUC 1.00 (
- XGBoost multiclass + physics ased split; inverse-frequency weighting
- Baseline profiler (statistical + I ; ms/event streaming scorer

Detection Architecture — Every Stage Writes Auditable Outputs



FEASIBILITY AND VIABILITY

- Analysis of the feasibility of the idea

Feasibility — proven, not promised:

- Working end-to-end pipeline on 73,931 sessions; live scoring at 3.6 ms/event (277 events/sec, single thread)
- PR-AUC 0.996 at 1:42 imbalance; 0 false alarms/day at a top-1% analyst budget; 50,271 events/sec batched

Challenges & risks — identified honestly:

- Results are measured on SYNTHETIC traffic and are optimistic by nature — 3 inflating artifacts were found and removed during development
- Measured on purpose-built cohorts: 20 brand-new entities (2,647 sessions) + 18 entities that legitimately changed shift and device

Mitigation strategies:

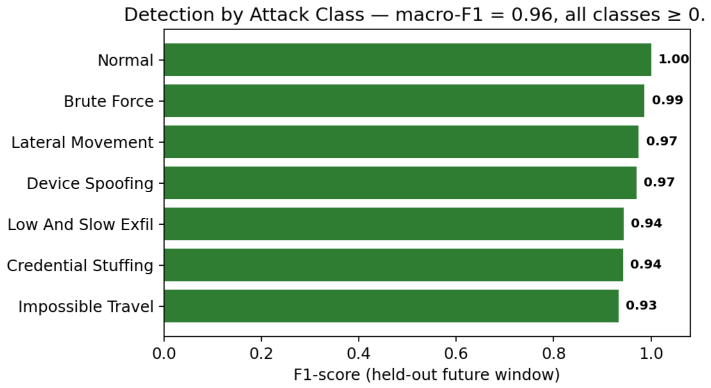
- 0.00% false positives on both cohorts; rolling baselines re-learn the new normal (post-drift deviation 0.85, identical to unchanged entities)
- Physics rule layer confirms AND vetoes impossible travel → that class improved 0.83 → 0.93 F1
- Streaming demonstrated by per-event benchmark; causal features port directly to Kafka-style stateful processing

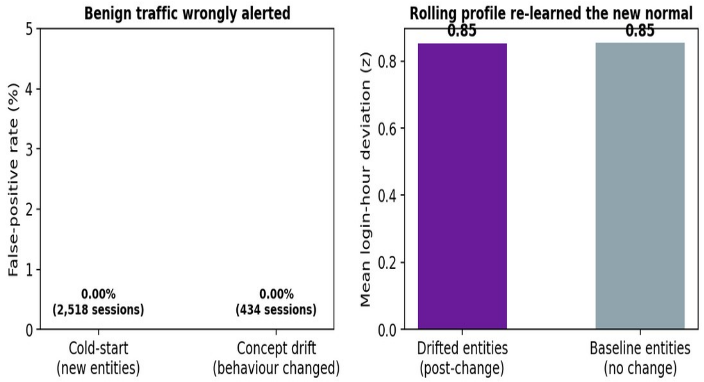
ARTIFACTS

- Relev
- GitHub repo
- Live outputs

Ranked Alert Queue — Analyst Console (live output)

Entity	Predicted Attack	Risk	Explainable Rationale (SHAP)
user_0048	Credential Stuffing	1.0	Flagged due to: distinct accounts from same source IP (1h) (10.0 accounts, S...
user_0022	Low And Slow Exfil	1.0	Flagged due to: number of privileged actions in session (1.0 actions, SHAP +...
user_0062	Lateral Movement	1.0	Flagged due to: graph_edge_weight (SHAP +3.63); graph_anomaly_score (SHAP +0...
user_0081	Brute Force	1.0	Flagged due to: time since entity's last session (0.1 min, SHAP +3.63); fail...
user_0013	Credential Stuffing	1.0	Flagged due to: distinct accounts from same source IP (1h) (12.0 accounts, S...
user_0038	Credential Stuffing	1.0	Flagged due to: distinct accounts from same source IP (1h) (5.0 accounts, SH...





RESEARCH AND REFERENCES

- Details / Links of the reference and research work

- Lundberg & Lee (2017), “A Unified Approach to Interpreting Model Predictions” (SHAP) — NeurIPS
- Chen & Guestrin (2016), “XGBoost: A Scalable Tree Boosting System” — KDD
- Chandola, Banerjee & Kumar (2009), “Anomaly Detection: A Survey” — ACM Computing Surveys
- MITRE ATT&CK® framework — attack taxonomy reference for injected patterns (attack.mitre.org)
- CERT Insider Threat studies — basis for low-and-slow and insider-drift simulation design

- **GitHub repository link: [to be added on submission portal]**